

CAPSTONE · SUPPLY CHAIN ANALYTICS

DataCo Supply Chain Performance Analysis

Shipping mode profitability and delivery performance

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Capstone Project

Business context and dataset overview

Nine connected datasets

Dataset	Records	Main role
Customers	20,652	Customer details and segments
Orders	65,752	Orders and delivery measures
Order Items	180,519	Sales, discounts and profit
Products	118	Product reference
Categories	51	Product categories
Departments	11	Department reference
Shipping	65,752	Shipping mode and dates
Geography	64,990	Markets and locations
Payment Types	4	Payment reference

DataCo’s supply chain connects customers, products and payments with shipping operations across geographic markets.

The dataset captures the order fulfillment journey, from customer orders and product purchases to shipping and delivery.

It supports order management, customer behavior analysis, product performance evaluation, logistics optimization and profitability reporting.

TAKEAWAY

Project focus: compare shipping modes using revenue, discounts, recorded profitability and delivery performance.

Problem statement: shipping performance and profit

How can DataCo compare sales, discounts, profit and delivery performance across its four shipping modes to find profitable options and suggest ways to reduce low-margin shipments?

Standard Class **Second Class** **First Class** **Same Day**

Focus: compare sales, discounts, profit margins and delivery performance for each shipping mode.

TAKEAWAY

Data limitation: separate delivery and handling costs are unavailable. We compare the profit and delivery results recorded in the data.

Five research objectives

01 Analyze Overall Sales Performance

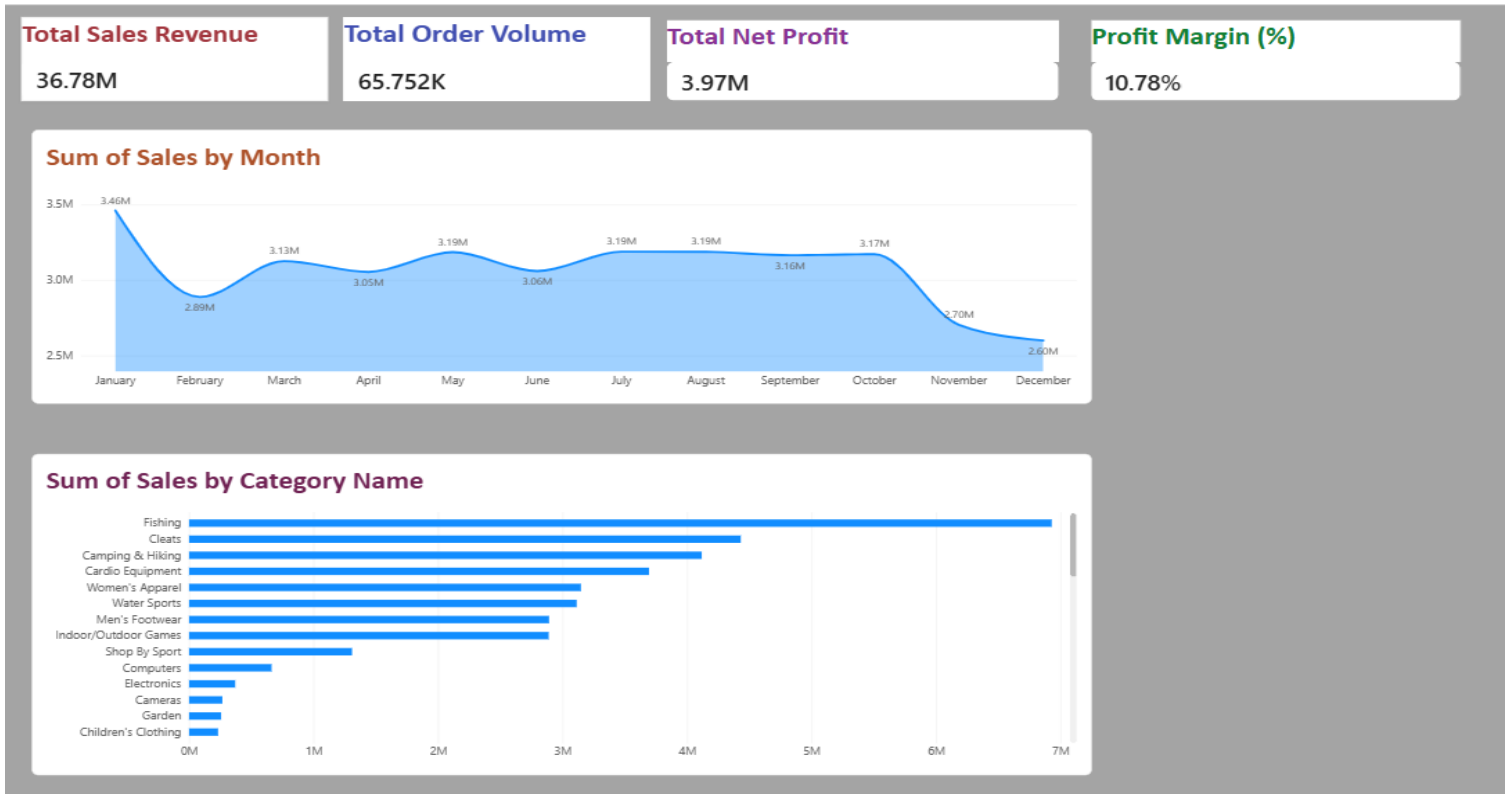
02 Evaluate Shipping Mode Performance & Fulfillment Efficiency

03 Analyze Product, Category and Department Profitability

04 Analyze Shipping Mode Profitability & Delivery Performance

05 Analyze Customer Segment Performance & Purchasing Patterns

Objective 1: Analyze Overall Sales Performance



Performance & core insight

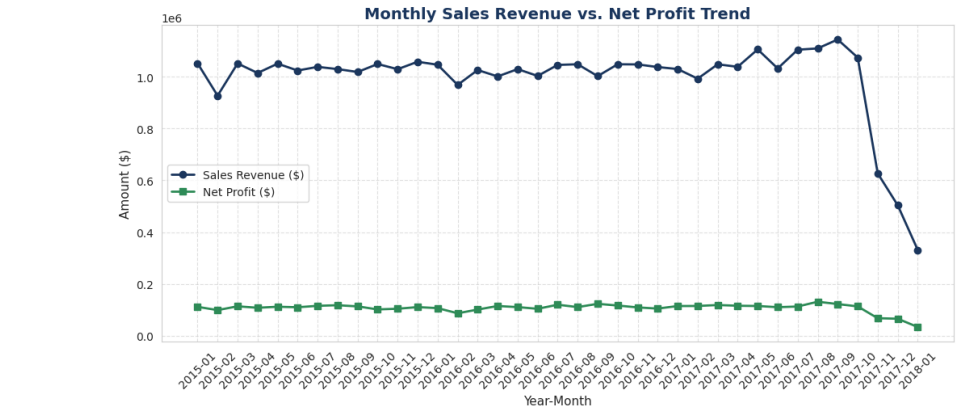
Revenue: USD 36.78M
Unique orders: 65,752
Recorded profit: USD 3.97M
Profit margin: 10.78%

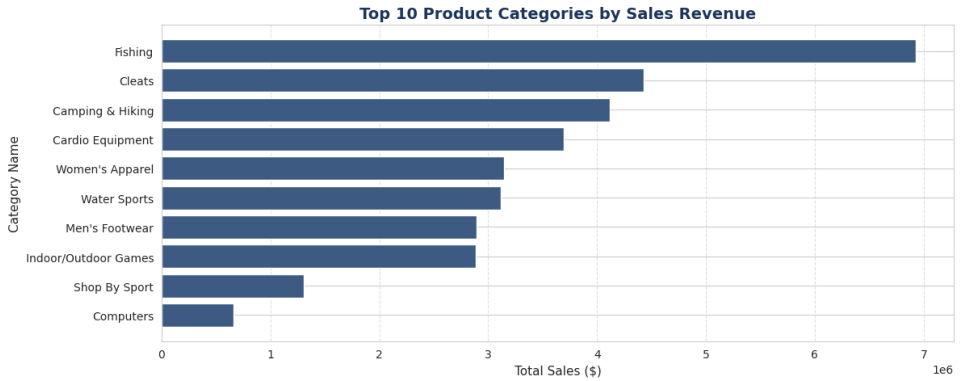
Core insight
About USD 10.78 recorded profit per USD 100 of sales. Separate fulfillment costs are unavailable.

TAKEAWAY

$36.78M sales 65,752 orders $3.97M reported profit 10.78% margin

Objective 1: Python sales trends and categories





Observation

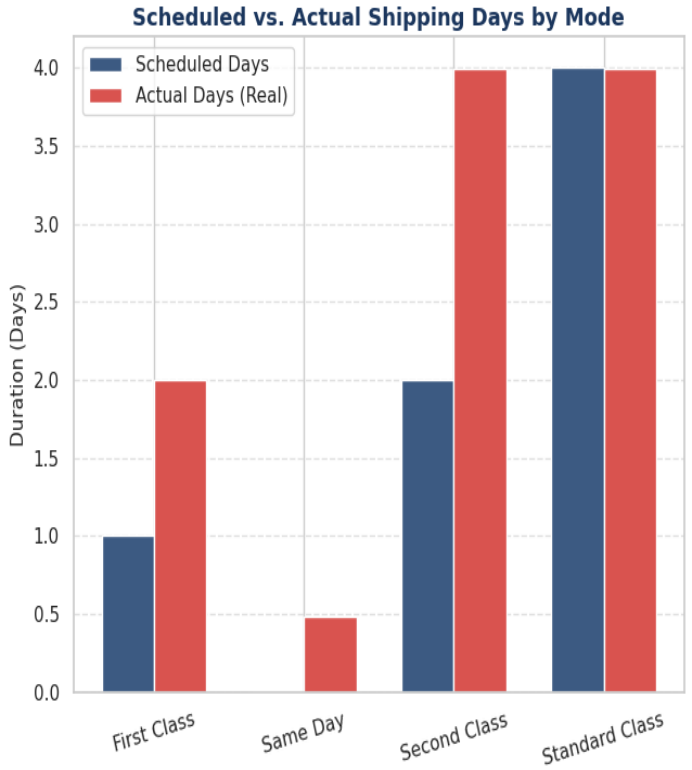
Fishing leads category sales at about $6.9M.

Sales decline at the end of the timeline. Check period coverage first.

TAKEAWAY

Recommendation: protect high-revenue categories and review low-margin items to improve overall profitability.

Objective 2: Shipping Mode Performance & Fulfillment Efficiency





Observation

Standard Class: about 4 days, matching its schedule. Late risk: 38.1%.

Same Day: about 0.5 day on average. Late risk: 45.7%.

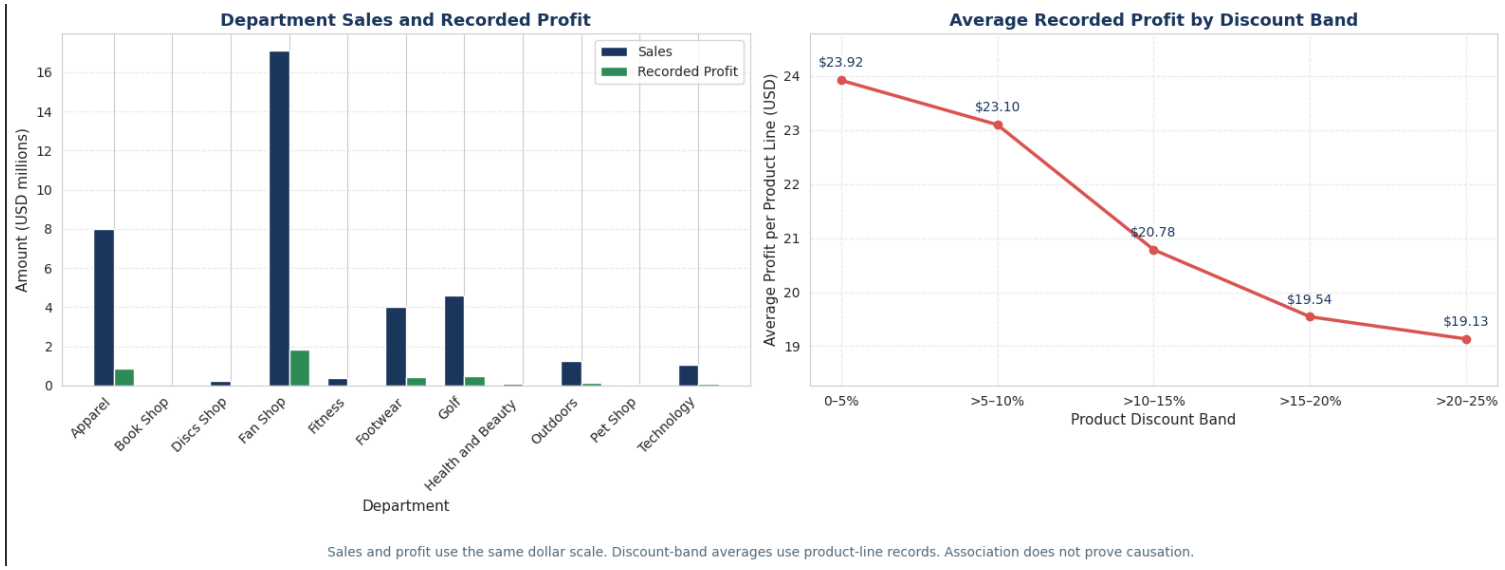
Second Class: about 4 days vs 2 scheduled. Late risk: 76.6%.

First Class: 2 days vs 1 scheduled. Late risk: 95.3%.

TAKEAWAY

Standard Class best matches its average schedule. Review First and Second Class delivery commitments with carriers.

Objective 3: Product, Category and Department Profitability



Sales and profit use the same dollar scale. Discount-band averages use product-line records. Association does not prove causation.

Performance & core insight

Fan Shop leads with about USD 17.11M sales and USD 1.83M recorded profit.

Average profit per product line falls from USD 23.92 at 0–5% discount to USD 19.13 above 20–25%.

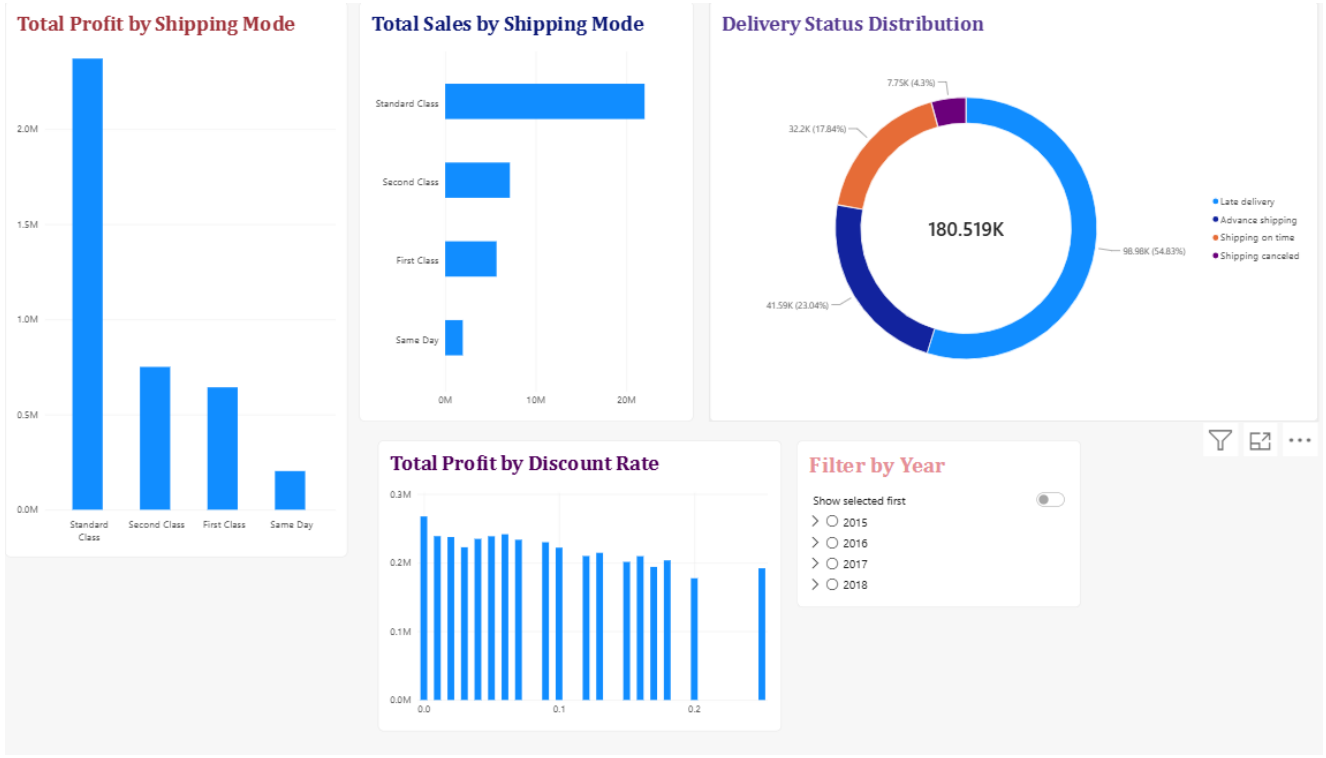
Higher discounts are linked to lower average profit.

TAKEAWAY

Recommendation: review low-margin products and test discount controls by shipping mode. Association does not prove causation.

Objective 4: Shipping Mode Profitability & Delivery Performance

Power BI shipping dashboard



Compare revenue and profit margins across shipping modes.

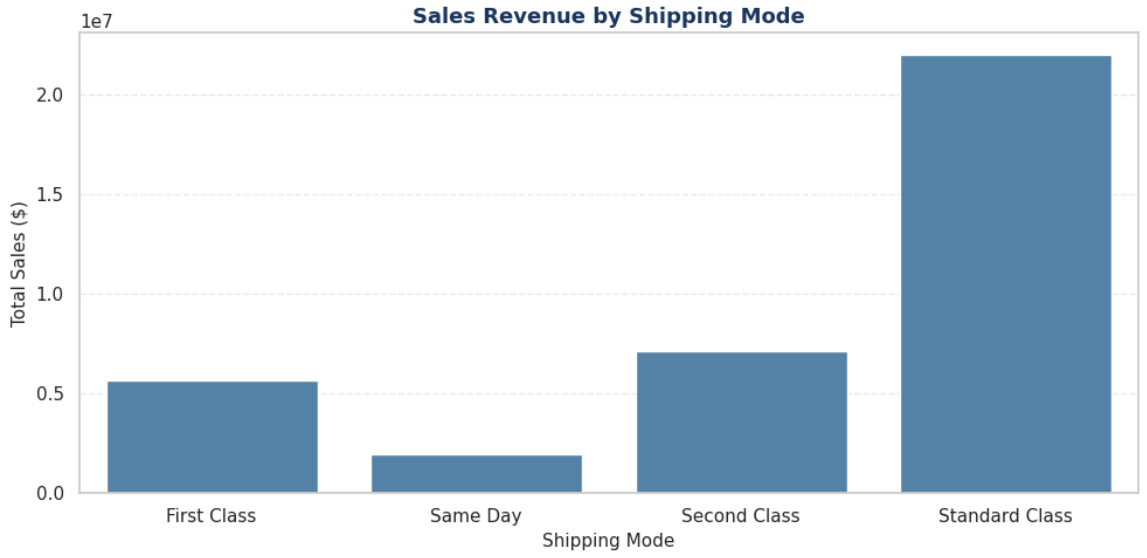
Evaluate recorded profitability alongside delivery performance.

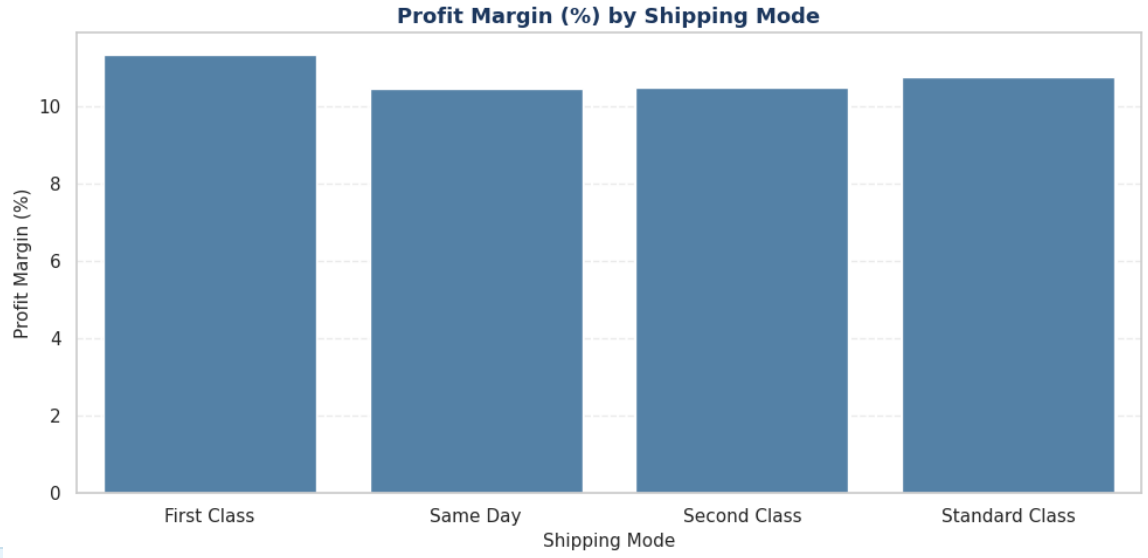
Build a predictive model to classify profitable and unprofitable shipments.

TAKEAWAY

Full dataset: late delivery accounts for 54.83% of 180,519 order-item records. Standard Class leads sales and total profit.

Objective 4: Python shipping profitability





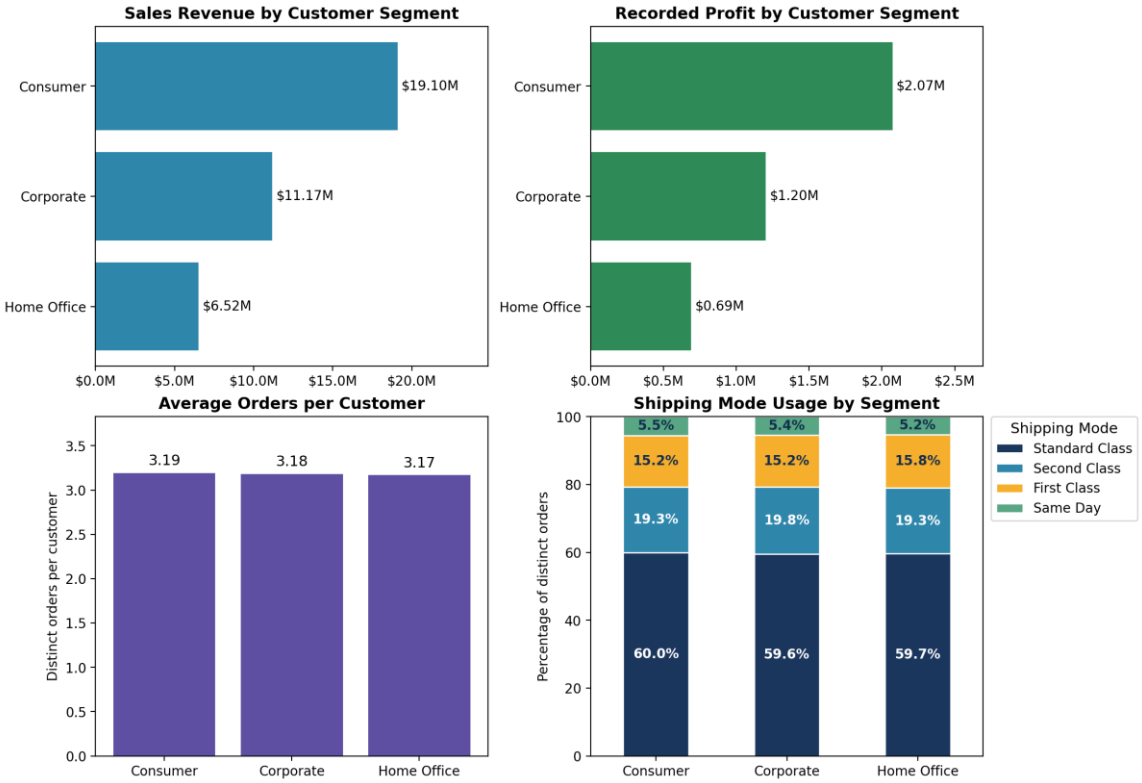
Performance & core insight

Standard Class: USD 22.02M sales and USD 2.37M profit. First Class leads margin at 11.33%, followed by Standard 10.76%, Second 10.50% and Same Day 10.45%. Volume and margin identify different leaders.

TAKEAWAY

Recommendation: protect Standard Class capacity and review premium service reliability. Collect fulfillment costs for future analysis.

Objective 5: Customer Segment Performance & Purchasing Patterns



Observation

1. Sales contribution
Consumer: USD 19.10M (52%). Corporate: USD 11.17M. Home Office: USD 6.52M.

2. Profit contribution
Consumer: USD 2.07M. Corporate: USD 1.20M. Home Office: USD 0.69M.

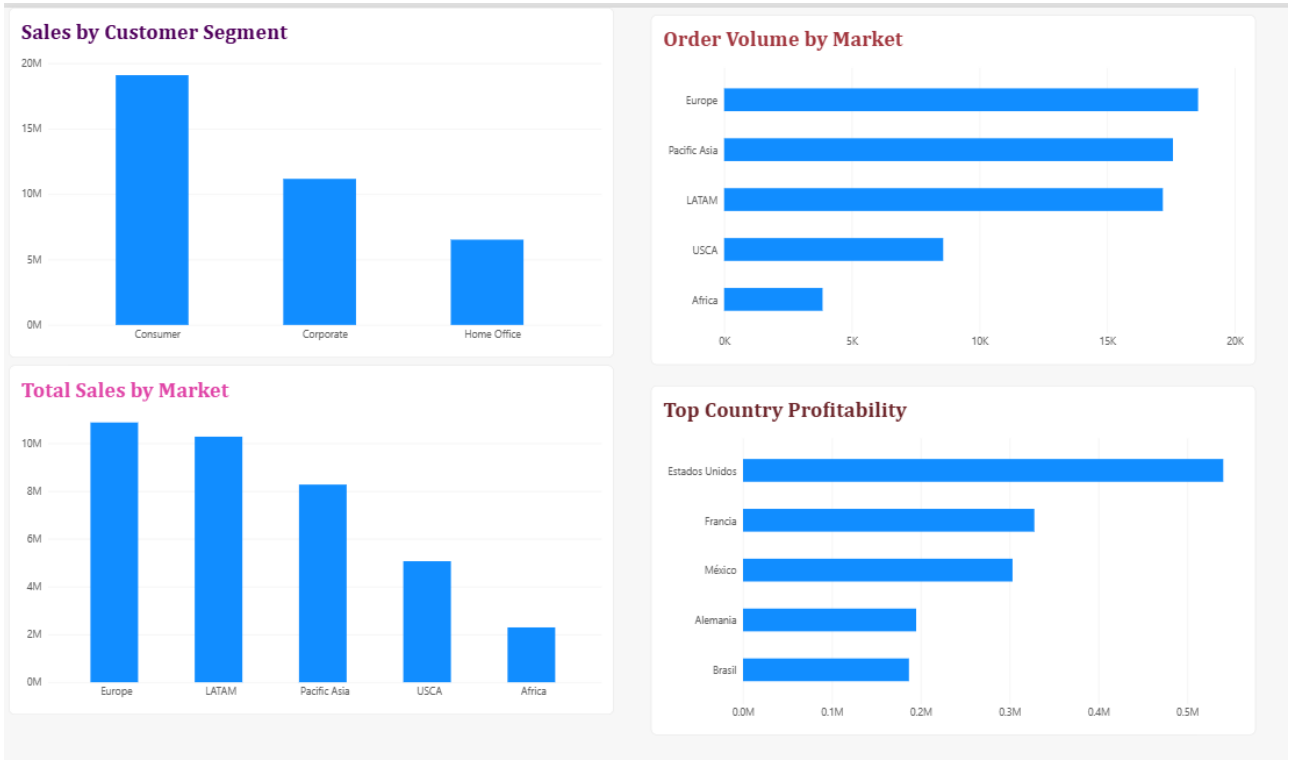
3. Purchasing patterns
Orders per customer: 3.19, 3.18 and 3.17. Frequency is similar across segments.

4. Shipping mode usage
Standard Class represents about 60% of orders in every segment.

TAKEAWAY

Core insight: Consumer drives scale, while ordering and shipping patterns are similar. Recommendation: maintain reliable Standard Class delivery and monitor profit across segments.

Objective 5: Power BI customer dashboard



TAKEAWAY

Consumer leads sales. Europe and LATAM lead market revenue.

Recommendation: support Consumer demand with reliable Standard Class service. Usage does not prove personal preference.

Full-dataset comparison of shipping modes

All years in the master dataset. Currency: USD.

Shipping mode	Scheduled days	Average actual days	Late-risk rate	Sales revenue	Recorded profit	Recorded margin
Standard Class	4	4.00	38.1%	$22.02M	$2.37M	10.76%
Second Class	2	3.99	76.6%	$7.15M	$750,308	10.50%
First Class	1	2.00	95.3%	$5.67M	$643,122	11.33%
Same Day	0	0.48	45.7%	$1.94M	$203,018	10.45%

Standard Class leads total profit and has the lowest late-risk rate.
First Class leads margin, but averages 2 days against 1 day scheduled.

Delivery averages and late-risk rates use order-item records. 0 scheduled days means same-day delivery.
Actual fulfillment charges are not separately available. Values are rounded.

Business recommendations

Evidence	Recommended action
First and Second Class miss average schedules	Review service promises and delay patterns
Average profit falls across discount bands	Pilot discount approval thresholds
Standard Class leads total profit	Protect capacity and monitor its service level
First Class leads margin but misses its schedule	Balance margin against premium service reliability
Actual fulfillment cost data is unavailable	Add carrier charges and handling costs by mode

Conclusion

DataCo recorded USD 36.78M in sales and USD 3.97M in profit, with a 10.78% profit margin.

Standard Class earns the most total profit. First Class has the highest margin, but takes 2 days on average against 1 day scheduled.

Higher discounts are linked to lower average profit. Discounts and low-margin sales need careful review.

Consumer contributes about 52% of sales. Standard Class serves about 60% of orders in every customer segment.

Main conclusion: balance profit and reliable delivery when choosing shipping modes.

Thank you · Questions and discussion